ARJUN PULIVARTHI

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PROFILE

Highly motivated and solutions-driven Computer Science undergraduate specializing in Artificial Intelligence and Machine Learning. Proven expertise in building real-world applications across financial forecasting, cybersecurity, recommendation systems, and IoT-based payment solutions. Co-founder of UniGrid, with demonstrated leadership in ideating, developing, and pitching scalable digital products. Adept in Python, Flask, SQL, and ML libraries, with experience in Explainable AI (XAI), time series modeling, and NLP. Passionate about solving impactful problems through innovation, with a strong portfolio of hackathon-winning projects and client-facing prototypes.

ACADEMIC QUALIFICATION

Bachelor of Technology in Computer Science (Artificial Intelligence and Machine Learning) ---------- Aug 2022 - Present

Dayananda Sagar University, Bengaluru, Karnataka.

Aggregate CGPA: 6.0 (Awaiting 6th Semester Results)

CBSE 12th ---------- May 2022

Sri Chaitanya Techno School, Bengaluru, Karnataka.

Percentage: 81%

CBSE 10th ---------- Mar 2020

Presidency School Bangalore South, Bengaluru, Karnataka.

Percentage: 76%

STARTUP

UniGrid – RFID-Enabled Smart Student ID and Payment System ---------- Jan 2025 - Present

Built RFID/NFC-enabled campus payment & ID system (UniGrid) with admin, cashier, and student panels. Integrated Razorpay for top-ups, Twilio SMS alerts, and receipt printing. Implemented role-based access, real-time balances, and ERP-ready sync. Leading pilot implementation in Indian educational institutions.

KEY PROJECTS

Semantic Similarity Based Academic Answer Evaluator using ML and BI ---------- Present

Capstone Project | Team Size: 4

Tech Stack: Python, NLTK, spaCy, Scikit-learn, TensorFlow, SBERT, Streamlit, Power BI

1. Engineered an AI-driven evaluator to grade descriptive answers using hybrid TF-IDF & SBERT semantic similarity models
2. Integrated plagiarism detection, Bloom's Taxonomy classification, and Explainable AI for transparent grading
3. Implemented bias detection & consistency analysis ensuring fairness across evaluations
4. Developed Power BI dashboards delivering actionable academic insights and performance analytics

Stock Market Price Predictor Using AI ---------- April 2025

Course Minor Project | Team Size: 1

Tech Stack: Python, Pandas, Scikit-learn, XGBoost, LSTM, Matplotlib, Streamlit

1. Built an AI-driven forecasting tool using XGBoost & LSTM on historical stock data
2. Implemented feature engineering for short term price prediction.
3. Evaluated models with moving averages, RSI, and volume trends
4. Developed interactive real-time dashboard using Streamlit for predictions & visualization

AI-Based Defect Detection in Ultrasonic Polymer Welding ---------- Dec 2024

Course Minor Project | Team Size: 4

Tech Stack: Python, OpenCV, NumPy, Scikit-learn, Matplotlib

1. Designed NDE image-processing pipeline for polymer weld defect detection
2. Applied advanced filtering, edge detection & adaptive thresholding for noise reduction
3. Used texture analysis (GLCM, LBP, entropy) achieving 93% classification accuracy
4. Research accepted at NDE 4.0 Conference; submitted to Springer Journal of NDE

Deepfake Detection Software Using Deep Learning ---------- Nov 2024

Personal Project | Team Size: 1

Tech Stack: Python, TensorFlow, Keras, OpenCV, Flask, PIL, Matplotlib

1. Built CNN-based tool to detect manipulated facial images
2. Preprocessed data with face detection & augmentation for training
3. Integrated heatmap visualizations to highlight forged regions
4. Deployed Flask-based interface; achieved high precision on benchmark datasets

Network Threat Identifier Using AI ---------- **Nov 2024**
Hackathon Project | PES University - AI/ML in Cybersecurity 2025 - 5th Place | Team Size: 4
Tech Stack: Python, Flask, Scapy, MongoDB, TensorFlow, CNN, Apache Kafka

1. Designed real-time intrusion detection using CNNs for packet classification
2. Implemented three-node architecture for monitoring & automatic recovery
3. Integrated fault-tolerant backup/sync mechanism after anomalies
4. Built live dashboards & alerts for proactive security monitoring

Gesture Articulation and Recognition ---------- **Jun 2024**
Personal Project | Team Size: 1
Tech Stack: Python, OpenCV, MediaPipe, NumPy, TensorFlow

1. Developed real-time hand gesture recognition system using MediaPipe + TensorFlow
2. Classified static poses (e.g., thumbs up) & dynamic gestures (e.g., swipe, wave)
3. Provided instant visual overlays for gesture tracking and debugging
4. Targeted applications in sign language, virtual controls, and HCI systems

Urban Forge Solutions - AI Toolkit for Urban Planning ---------- **Feb 2024**
Hackathon Project | RV Hack4Soc 2.0 | Recognized for Innovation | Team Size: 4
Tech Stack: Python, Flask, OpenCV, Scikit-learn, Google Maps API, JavaScript, HTML/CSS, Dialogflow

1. Built AI toolkit with floor plan generation, plot price prediction & chatbot support
2. Designed location-based pricing model via Google Maps API
3. Integrated Dialogflow chatbot for urban layout queries & negotiation support
4. Recognized for **innovation** at RV Hack4Soc 2.0 Hackathon

IoT-Based AI Farming System ---------- **Dec 2023**
Hackathon Project | Reva Smart Agro Hackathon 2023 - 3rd Place | Team Size: 4
Tech Stack: Python, Arduino, NodeMCU, TensorFlow, Firebase, Dialogflow, OpenWeatherMap API, IoT Sensors, Google Speech API

1. Created smart farming system recommending crops via land & environmental data
2. Automated irrigation using NodeMCU-controlled actuators with soil sensors
3. Integrated real-time weather API for predictive irrigation scheduling
4. Built multilingual, voice-enabled chatbot for farmer guidance (Dialogflow + Google Speech API)
5. Secured 3rd Place at Reva Smart Agro Hackathon 2023

Virtual Meeting Analysis Platform (Insight Flow) ---------- **Oct 2023**
Hackathon Project | NIT Surathkal | Team Size: 4
Tech Stack: Python, Flask, HatBERT (custom RoBERTa-based NLP model), NLTK, WebSockets, HTML, CSS, JavaScript, REST APIs, Google Cloud, Kaggle

1. Developed AI platform to analyze team roles in meetings using Six Thinking Hats model
2. Created HatBERT (RoBERTa-based NLP classifier) for role & sentiment detection
3. Enabled live annotation via WebSocket infrastructure for real-time tracking
4. Built interactive UI to enhance SME team collaboration and awareness

Sakhi Samriddhi – AI-Based Menstrual Health Prediction and Chatbot System ---------- **Nov 2023**
Hackathon Project | KLE GirlGeekHack 2023 - 1st Place | Team Size: 4
Tech Stack: Python, TensorFlow, Flask, Dialogflow, HTML/CSS, Bootstrap

1. Built ML model to forecast menstrual cycles using user health data
2. Developed AI chatbot (Dialogflow) for health alerts & personalized guidance
3. Integrated model outputs with web-based interface for easy access
4. Won 1st Place at KLE GirlGeekHack 2023 for women's health innovation

Cafe Management System – GUI-Based Billing and Order Application ---------- **Mar 2022**
Course Project | Team Size: 2
Tech Stack: Python, Tkinter, MySQL

1. Developed Python desktop app for café order management & tax-inclusive billing
2. Designed GUI using Tkinter for real-time item selection & receipt generation
3. Integrated MySQL for managing menu items, orders & transaction logs

Timetable Maker - Automated Class Schedule Generator ---------- **Oct 2019**
Personal Project | Team Size: 1
Tech Stack: Python

1. Built Python-based tool to auto-generate weekly school timetables
2. Implemented constraints to avoid subject repetition & ensure balanced schedules
3. Designed to reduce manual effort and errors in timetable planning

FREELANCE PROJECTS

AI IPL Win Predictor - Real-Time Cricket Match Outcome Forecasting ---------- **Jan 2025**

Tech Stack: Python, Pandas, Scikit-learn, Streamlit, Matplotlib

1. Developed logistic regression model to predict IPL match outcomes from ball-by-ball data
2. Engineered features like run rate, wickets left, overs remaining & momentum factors
3. Built dynamic Streamlit dashboard updating win probabilities after every delivery
4. Integrated interactive visualizations for momentum shifts & statistical insights
5. Delivered as a freelance solution for fantasy league analytics & live match dashboards

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LEADERSHIP & EVENT MANAGEMENT

Chairperson, IEEE Student Branch ---------- **Jan 2024 – Jan 2025**

Dayananda Sagar University, Bengaluru, India

Built the IEEE Student Community in my University. Led international tech conferences, speaker sessions, and innovation forums under the IEEE banner. Oversaw strategic planning, member engagement, and collaboration with global IEEE chapters. Spearheaded initiatives to enhance student participation in AI/ML-focused competitions and research workshops.

Chairperson, IEEE CIS Student Chapter ---------- **Dec 2023 – Apr 2024**

Dayananda Sagar University, Bengaluru, India

Organized AI/ML technical workshops, coding competitions, and expert lectures across the university. Managed content curation, logistics, and cross-departmental coordination for high-impact events. Fostered collaboration among student chapters through joint events and knowledge-sharing initiatives.

Organizational Lead ---------- **Mar 2024**

ICIQ-24 Hackathon – DSU x LeTourneau University (USA)

Served as core organizing member for a 30-hour international hackathon involving 60+ student teams and 200+ stakeholders. Handled end-to-end logistics including registration, team allocation, tech infrastructure, and live support coordination. Collaborated with global delegates from Google, ISRO, and US universities to ensure smooth cross-time-zone participation. Contributed to content creation, marketing workflows, and real-time crisis resolution during the event.

Disciplinary Committee Head ---------- **Oct 2024 – Present**

Centre for Performing Arts (CPA), Dayananda Sagar University

Led discipline, safety, and backstage coordination for high-footfall college fests and cultural events. Enforced protocol across event zones, collaborated with volunteers and university staff, and ensured smooth event operations.

Volunteer & Coordinator ---------- **Dec 2023**

IEEE TEMSCON Asia-Pacific 2023, Bengaluru, India

Managed attendee registration and logistics for an international IEEE conference held at Sheraton Grande, Bengaluru. Facilitated smooth execution of sessions, handled speaker coordination, and supported backend operations. Collaborated with senior IEEE professionals to coordinate tracks across technology, engineering, and innovation domains. Assisted in managing panel discussions, Q&A sessions, and emergency protocol during high-attendance keynotes.

PUBLICATIONS

Unlocking Insights in Ultrasonic Polymer Welding: Advanced Image Processing and Data Integrity for SEM-Based Quality Analysis in NDE 4.0

Accepted at the 2nd International Conference & Exhibition on NDE 4.0 (Paper ID: CP-9, AB-010), 2024.

Semantic Similarity-Based Academic Descriptive Answer Evaluator Leveraging Machine Learning and Business Intelligence

Accepted at the MAHSA International Conference on Innovation in Research, Information, and Communication Technology (MiIRICT 2025), Malaysia.

WORK EXPERIENCE

Research Intern ---------- **May 2025 - Present**

National Institute of Technology Puducherry (NIT-PY), Karaikal, India

Assisted in the development of AI/ML-based systems as part of an academic research initiative. Contributing to literature review, data preprocessing, and implementation of machine learning models for heavy vehicles and collaborated with research mentors and graduate students on publication-focused outcomes.

Chief Technical Officer (CTO) & Engineering Lead ---------- **Jan 2025 – Present**

UniGrid Startup, Bengaluru, India

Lead architect of backend systems, ensuring platform scalability, security, and fault-tolerant design using Flask, MongoDB, and SQLite. Led development of integrated Parent, Teacher, and Student apps, with modules for performance monitoring, leave approval, and real-time alerts. Recruited and managed technical team members; established sprint cycles, infrastructure plans, and product roadmaps. Oversaw end-to-end deployment for MVP pilot at PUC-level institutions, focusing on usability, institutional feedback loops, and analytics dashboards.

AWARDS & ACHIEVEMENTS

1. **5th Place**, PES University AI/ML in Cybersecurity Hackathon 2025
2. **Best Innovation Mention**, RV Hackathon Hack4Soc 2.0, Feb 2024
3. **Lead Organizer**, ICIQ-24 Hackathon, Mar 2024
4. **3rd Place**, Reva Smart Agro Hackathon 2023, Dec 2023
5. **1st Place**, KLE College GirlGeekHack 2023, Nov 2023
6. **Appointed Chairperson**, IEEE Student Branch & IEEE CIS Chapter, 2023–2025

SKILLS

Programming Languages ----------
Python (expert), Java (academic), C (expert), HTML, CSS, JavaScript, SQL

Frameworks & Libraries ----------
Flask, Streamlit, Tkinter, Bootstrap, OpenCV, MediaPipe, TensorFlow, Keras, Scikit-learn, Numpy, Pandas, Matplotlib, Seaborn

Machine Learning & AI ----------
Convolutional Neural Networks (CNN), LSTM, XGBoost, Logistic Regression, HatBERT, Natural Language Processing (NLTK, Dialogflow), Recommendation Systems, Time Series Forecasting, Explainable AI (SHAP & LIME), Feature Engineering, Image Segmentation, Thresholding, Texture Analysis (GLCM, LBP), Entropy-based Detection

Tools & Platforms ----------
Git, GitHub, Jupyter Notebook, PyCharm, VS Code, Arduino IDE, Firebase, Google Cloud Platform, MongoDB, REST APIs, WebSockets, Apache Kafka, Scapy, MySQL, SQLite

Data Visualization & Analysis ----------
Matplotlib, Seaborn, Excel, Streamlit Dashboards, Heatmaps, Confusion Matrices

Hardware & Embedded Systems ----------
Arduino, NodeMCU, RFID/NFC, Thermal Printer Integration, IoT Sensors

Soft Skills & Leadership ----------
Team Management, Hackathon Organization, Cross-Functional Collaboration, Technical Communication, Strategic Planning, Public Speaking, Mentoring Peers, Client Interaction

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CERTIFICATIONS & COURSES

Deep Learning and Neural Networks ---------- **Sep 2025**
Udemy Certificate ID: UC-a2818ead-adc4-446e-802a-ff86ae8eca61
Studied CNNs, RNNs, transformers and built projects applying deep learning to real cases.

Complete Data Science and Machine Learning Bootcamp ---------- **Sep 2025**
Udemy Certificate ID: UC-232f725d-9d34-45d7-aa99-33b718ba2213
Learnt end-to-end ML, data preprocessing, supervised/unsupervised models, deep learning and deployment basics.

FOSSMeet '24 ---------- **Feb 2024**
NIT Calicut, Kerala, India
Participated in sessions on Free and Open-Source Software, developer best practices, community-driven tech, and open AI tooling.

Signals and Systems ---------- **Dec 2023**
Udemy Certificate ID: UC-63ef1c59-0a56-40a2-b7c3-e6e693609822
Studied signal processing fundamentals, Fourier series, Laplace transforms, and system analysis for AI/ML signal inputs.

Data Structures and Algorithms ---------- **Nov 2022**
Udemy Certificate ID: UC-9b39c8c8-8bd4-4df2-a4dc-8bd3c94bc033
Learned sorting/searching algorithms, recursion, trees, graphs, and problem-solving patterns for coding interviews.

Complete Python Bootcamp ---------- **Sep 2022**
Udemy Certificate ID: UC-80c5ada9-bc51-4184-aeaa-04c9520f0a93
Covered Python fundamentals, data structures, OOP, file handling, web scraping, and introductory data visualization.

I hereby declare that the information provided above is true and correct to the best of my knowledge and belief.
Place: Bengaluru

Arjun Pulivarthi